

Solution to Exercise 08 — Analysis on Manifolds, 80416

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Question 1

Subquestion 1

We will use Guldin's theorem to calculate the area of $S^2 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$.

Solution: We follow the recitation's definition, let $\gamma : (a, b) \rightarrow \mathbb{R}^2$ be a smooth map denoted by $\gamma(t) = (x(t), z(t))$.

We need to ensure that $x(t) > 0$ for all $t \in (a, b)$ so $x(t) = \cos(t)$, $z(t) = \sin(t)$ for $t \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

Our surface of revolution is the sphere and it's obtained by γ as

$$M = \left\{ (x(t) \cos(\theta), x(t) \sin(\theta), z(t)) \mid t \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \theta \in [0, 2\pi] \right\}$$

we have

$$\|\dot{\gamma}(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = \sqrt{\sin^2(t) + \cos^2(t)} = \sqrt{1} = 1$$

Hence

$$l(\gamma) = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \|\dot{\gamma}(t)\| dt = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 1 dt = \pi$$

Now we need to find the center of mass a_1 .

Let a be the center of mass of $\gamma(t)$ in $\mathbb{R}^3$ so

$$a_1 = \frac{1}{l(\gamma)} \int_{\gamma} x d\sigma$$

We know that

$$\int_{\gamma} x d\sigma = \int_a^b x(t) \|\dot{\gamma}(t)\| dt$$

Hence

$$a_1 = \frac{1}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(t) dt = \frac{1}{\pi} \cdot [\sin(t)]_{t=-\frac{\pi}{2}}^{t=\frac{\pi}{2}} = \frac{1}{\pi} (1 - (-1)) = \frac{2}{\pi}$$

Now we apply Guldin's theorem

$$\text{Area}(M) = l(\gamma) \cdot 2\pi |a_1|$$

So in our case

$$\text{Area}(M) = \pi \cdot 2\pi \cdot \frac{2}{\pi} = 4\pi$$

□

Subquestion 2

We will use Guldin's theorem to calculate the area of $C = \{(x, y, z) \in \mathbb{R}^3 \mid z^2 = a^2(x^2 + y^2), 0 < z < h, a, h > 0\}$.

Solution: To find the generating curve γ in the xz -plane we set $y = 0$ so from the equation of the cone we have $z^2 = a^2 x^2$.

Since $a > 0$ and $x(t) > 0$ we can take root and get that $z = ax$.

This is a line segment that can be parametrized as $x(t) = t$, $z(t) = at$ and we have constraint on z with h so

$$0 < z < h \iff 0 < at < h \implies 0 < t < \frac{h}{a}$$

So we define $\gamma : (0, \frac{h}{a}) \rightarrow \mathbb{R}^2$ to be the smooth map $\gamma(t) = (x(t), z(t))$ with $x(t) = t$, $z(t) = at$ and $t \in (0, \frac{h}{a})$.

So we can define

$$M = \left\{ (t \cos(\theta), t \sin(\theta), at) \mid t \in \left(0, \frac{h}{a}\right), \theta \in [0, 2\pi] \right\}$$

Same as before

$$\dot{\gamma}(t) = (x'(t), z'(t)) = (1, a)$$

$$\|\dot{\gamma}(t)\| = \sqrt{1^2 + a^2} = \sqrt{1 + a^2}$$

$$l(\gamma) = \int_0^{\frac{h}{a}} \|\dot{\gamma}(t)\| dt = \int_0^{\frac{h}{a}} \sqrt{1 + a^2} dt = \sqrt{1 + a^2} [t]_{t=0}^{t=\frac{h}{a}} = \frac{h}{a} \sqrt{1 + a^2}$$

$$a_1 = \frac{1}{l(\gamma)} \int_{\gamma} x d\sigma = \frac{1}{l(\gamma)} \int_0^{\frac{h}{a}} t \|\dot{\gamma}(t)\| dt = \frac{1}{\frac{h}{a} \sqrt{1 + a^2}} \int_0^{\frac{h}{a}} t \sqrt{1 + a^2} dt = \frac{a}{2h} [t^2]_{t=0}^{t=\frac{h}{a}} = \frac{a}{h} \cdot \frac{h^2}{2a^2} = \frac{h}{2a}$$

Since $a, h > 0$ we have that $a_1 > 0$ so $|a_1| = a_1$ and from Guldin's theorem we get that

$$\text{Area}(M) = \left(\frac{h}{a} \sqrt{1 + a^2} \right) \cdot 2\pi \cdot \frac{h}{2a} = \frac{2\pi h^2 \sqrt{1 + a^2}}{2a^2} = \frac{\pi h^2 \sqrt{1 + a^2}}{a^2}$$

□

Question 2

Consider

$$M = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1, \sqrt{x^2 + y^2} \leq z^2\}$$

$$F(x, y, z) = \left(\frac{x^2}{2} + xy - 4xz + \cos(z), -\frac{y^2}{2} - xy + x^2 + z^3, 2z^2 + x \right)$$

We will calculate the flux of F through M with respect to the unit normal pointing upward.

Solution: Let $F = (P, Q, R)$ and we compute the divergence of F

$$P_x = \frac{\partial}{\partial x} \left(\frac{x^2}{2} + xy - 4xz + \cos(z) \right) = x + y - z$$

$$Q_y = \frac{\partial}{\partial y} \left(-\frac{y^2}{2} - xy + x^2 + z^3 \right) = -y - x$$

$$R_z = \frac{\partial}{\partial z} (2z^2 + x) = 4z$$

By definition

$$\operatorname{div} F = P_x + Q_y + R_z = (x + y - z) + (-y - x) + 4z = 0$$

We need to find the boundary of M .

Let $r = \sqrt{x^2 + y^2}$ and our conditions becomes $r^2 + z^2 = 1$ and $r \leq z^2$ so $r \leq 1 - r^2$ means that $r^2 + r - 1 \leq 0$ and since $r \geq 0$ possible range is $0 \leq r \leq \frac{\sqrt{5}-1}{2}$ so let $R = \frac{\sqrt{5}-1}{2}$.

On the boundary we have $z^2 = 1 - R^2 = R$ so $z = \sqrt{R}$ and $z = -\sqrt{R}$ thus M consists of two disjoint parts M_1 where $z > 0$ and M_2 where $z < 0$.

Same as the recitation, we denote

$$D_1 = \{(x, y, z) \in \mathbb{R}^3 \mid z = \sqrt{R}, x^2 + y^2 < R^2\}$$

$$V_1 = \{(x, y, z) \in \mathbb{R}^3 \mid \sqrt{R} < z < \sqrt{1 - x^2 - y^2}, x^2 + y^2 < R^2\}$$

$$D_2 = \{(x, y, z) \in \mathbb{R}^3 \mid z = -\sqrt{R}, x^2 + y^2 < R^2\}$$

$$V_2 = \{(x, y, z) \in \mathbb{R}^3 \mid -\sqrt{1 - x^2 - y^2} < z < -\sqrt{R}, x^2 + y^2 < R^2\}$$

By the divergence theorem

$$\int_{M_1 \cup D_1} \langle F, N \rangle d\sigma = \int_{M_1} \langle F, N \rangle d\sigma = \int_{D_1} \langle F, N \rangle d\sigma = \int_{V_1} \operatorname{div} F dx dy dz = 0$$

$$\int_{M_2 \cup D_2} \langle F, N \rangle d\sigma = \int_{M_2} \langle F, N \rangle d\sigma = \int_{D_2} \langle F, N \rangle d\sigma = \int_{V_2} \operatorname{div} F dx dy dz = 0$$

Let $N = (0, 0, -1)$. Let $p \in D_1$ and let $\gamma : (-\varepsilon, \varepsilon) \rightarrow D_1$ be a differentiable path such that $\gamma(0) = p$.

Since $\operatorname{Im} \gamma \subseteq D_1$ we can write it as $\gamma(t) = (x(t), y(t), \sqrt{R})$ thus

$$\langle \dot{\gamma}(0), N \rangle = \left\langle \begin{pmatrix} \dot{x}(0) \\ \dot{y}(0) \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} \right\rangle = 0$$

Moreover, for every $0 < s < 1$ we have

$$p + sN = (p_x, p_y, p_z - s) = (p_x, p_y, \sqrt{R} - s)$$

Where $p_z = \sqrt{R}$ since $p \in D_1$. For every $(x, y, z) \in V_1$ we have $z > \sqrt{R}$ hence $p + sN \notin V_1$ for all $0 < s < 1$.

Consider the map $\varphi : (0, R) \times (0, 2\pi) \rightarrow D_1$ defined by $\varphi(t, \theta) = (t \cos(\theta), t \sin(\theta), \sqrt{R})$ so we have

$$\begin{aligned}\frac{\partial \varphi}{\partial t} &= \begin{pmatrix} \cos(\theta) \\ \sin(\theta) \\ 0 \end{pmatrix}, & \frac{\partial \varphi}{\partial \theta} &= \begin{pmatrix} -t \sin(\theta) \\ t \cos(\theta) \\ 0 \end{pmatrix} \\ \Rightarrow \langle \frac{\partial \varphi}{\partial t}, \frac{\partial \varphi}{\partial \theta} \rangle &= -t \sin(\theta) \cos(\theta) + t \sin(\theta) \cos(\theta) = 0\end{aligned}$$

Furthermore

$$\begin{aligned}\left\| \frac{\partial \varphi}{\partial t} \right\|^2 &= \cos^2(\theta) + \sin^2(\theta) = 1 \\ \left\| \frac{\partial \varphi}{\partial \theta} \right\|^2 &= t^2 \cos^2(\theta) + t^2 \sin^2(\theta) = t^2\end{aligned}$$

Hence

$$\begin{aligned}\int_{D_1} \langle F, N \rangle d\sigma &= \int_{D_1} -(2z^2 + x) d\sigma \\ &= \int_0^{2\pi} \int_0^R -(2(\sqrt{R})^2 + t \cos(\theta)) \sqrt{t^2} dt d\theta \\ &= \int_0^{2\pi} \int_0^R -(2Rt + t^2 \cos(\theta)) dt d\theta \\ &= \int_0^{2\pi} \left[-Rt^2 - \frac{t^3}{3} \cos(\theta) \right]_{t=0}^{t=R} d\theta \\ &= \int_0^{2\pi} \left(-R^3 - \frac{R^3}{3} \cos(\theta) \right) d\theta \\ &= \left[-R^3 \theta - \frac{R^3}{3} \sin(\theta) \right]_{\theta=0}^{\theta=2\pi} \\ &= -2\pi R^3\end{aligned}$$

For M_1 the required normal N_{up} so

$$\int_{M_1} \langle F, N_{\text{up}} \rangle d\sigma = - \int_{D_1} \langle F, N \rangle d\sigma = 2\pi R^3$$

By identical reasoning for D_2 from V_2 the normal is $N = (0, 0, 1)$ and we have

$$\int_{D_2} \langle F, N \rangle d\sigma = \int_{D_2} (2z^2 + x) d\sigma = \int_0^{2\pi} \int_0^R (2(-\sqrt{R})^2 + t \cos(\theta)) t dt d\theta = \int_0^{2\pi} \left[Rt^2 + \frac{t^3}{3} \cos(\theta) \right]_{t=0}^{t=R} d\theta = 2\pi R^3$$

Hence

$$- \int_{M_2} \langle F, N_{\text{up}} \rangle d\sigma + 2\pi R^3 = 0 \Rightarrow \int_{M_2} \langle F, N_{\text{up}} \rangle d\sigma = 2\pi R^3$$

So the total upward flux is

$$\int_M \langle F, N_{\text{up}} \rangle d\sigma = \int_{M_1} \langle F, N_{\text{up}} \rangle d\sigma + \int_{M_2} \langle F, N_{\text{up}} \rangle d\sigma = 2\pi R^3 + 2\pi R^3 = 4\pi R^3$$

We have $R = \frac{\sqrt{5}-1}{2}$ as the positive root of $R^2 + R + 1 = 0$ so

$$R^3 = R(1 - R) = R - R^2 = R - (1 - R) = 2R - 1 = \sqrt{5} - 2$$

So the final flux is $4\pi(\sqrt{5} - 2)$. □

Question 3

Let $\Omega \subseteq \mathbb{R}^d$ be a bounded smooth open set. Let $U \subseteq \mathbb{R}^d$ be an open set such that $\overline{\Omega} \subseteq U$, let $u, v : U \rightarrow \mathbb{R}$ be smooth functions and let $X : U \rightarrow \mathbb{R}^d$ be a smooth vector field.

We will show that

$$\int_{\Omega} \Delta u \cdot \langle \nabla u, x \rangle dx = \int_{\partial\Omega} \frac{\partial u}{\partial N} \langle \nabla u, x \rangle d\sigma - \frac{1}{2} \int_{\partial\Omega} \|\nabla u\|^2 \langle x, N \rangle d\sigma + \frac{d-2}{2} \int_{\Omega} \|\nabla u\|^2 dx$$

Where N is the outward pointing unit normal of Ω and $\frac{\partial u}{\partial N} := \langle \nabla u, N \rangle$.

Proof: We set $\partial_i = \frac{\partial}{\partial x_i}$ so the laplacian is $\Delta u = \partial_{ii} u$ and the directional derivative along the position vector field is $\langle \nabla u, x \rangle = x_j \partial_j u$ and our integral on the right side becomes

$$\int_{\Omega} \Delta u \langle \nabla u, x \rangle dx = \int_{\Omega} (\partial_{ii} u)(x_j \partial_j u) dx$$

By the Divergence Theorem

$$\int_{\Omega} (\partial_{ii} u)(x_j \partial_j u) dx = \int_{\partial\Omega} N_i (\partial_i u)(x_j \partial_j u) d\sigma - \int_{\Omega} (\partial_i u) \partial_i (x_j \partial_j u) dx$$

We notice that

$$\begin{aligned} N_i \partial_i u &= \langle \nabla u, N \rangle = \frac{\partial u}{\partial N} \\ x_j \partial_j u &= \langle \nabla u, x \rangle \end{aligned}$$

This simplifies the first term with the boundary integral

$$(\star) \quad \int_{\Omega} (\partial_{ii} u)(x_j \partial_j u) dx = \int_{\partial\Omega} \frac{\partial u}{\partial N} \langle \nabla u, x \rangle d\sigma - \int_{\Omega} (\partial_i u) \partial_i (x_j \partial_j u) dx$$

Now, using the product rule we compute $\partial_i (x_j \partial_j u)$ we obtain that

$$\partial_i (x_j \partial_j u) = (\partial_i x_j)(\partial_j u) + x_j \partial_i (\partial_j u)$$

For smooth functions we have $\partial_i (\partial_j u) = \partial_{ij} u$ so we have

$$(\partial_i x_j)(\partial_j u) + x_j \partial_{ij} u$$

We use Kronecker Delta / The Dirac measure to see that

$$\delta_{ij} = \partial_i x_j = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

Hence

$$(\partial_i x_j)(\partial_j u) + x_j \partial_{ij} u = \delta_{ij} \partial_j u + x_j \partial_{ij} u = \partial_i u + x_j \partial_{ij} u$$

So

$$(\partial_i u)(\partial_i u + x_j \partial_{ij} u) = (\partial_i u)^2 + x_j (\partial_i u)(\partial_{ij} u) = \|\nabla u\|^2 + x_j (\partial_i u)(\partial_{ij} u)$$

We look at $x_j (\partial_i u)(\partial_{ij} u)$, if we think of $(\partial_i u)$ as a single function f this term looks like $f \cdot \frac{\partial f}{\partial x_j}$ and since $\frac{\partial}{\partial x_j} (f^2) = 2f \frac{\partial f}{\partial x_j}$ we get that $f \frac{\partial f}{\partial x_j} = \frac{1}{2} \frac{\partial}{\partial x_j} (f^2)$ so

$$(\partial_i u)(\partial_{ij} u) = \frac{1}{2} \partial_j ((\partial_i u)^2)$$

We need to sum both sides over i hence

$$\sum_{i=1}^d (\partial_i u)(\partial_{ij} u) = \frac{1}{2} \partial_j \sum_{i=1}^d (\partial_i u)^2 = \frac{1}{2} \partial_j (\|\nabla u\|^2)$$

So we have

$$-\int_{\Omega} [(\partial_i u)^2 + x_j(\partial_i u)(\partial_{ij} u)] dx = -\int_{\Omega} \|\nabla u\|^2 dx - \frac{1}{2} \int_{\Omega} x_j \partial_j (\|\nabla u\|^2) dx$$

We can apply the divergence theorem on the term $\int_{\Omega} x_j \partial_j (\|\nabla u\|^2) dx$ if we look at the vector field $F = \|\nabla u\|^2$ and the divergence of F is

$$\operatorname{div}(F) = \nabla(\|\nabla u\|^2 x) = \partial_j (\|u\|^2 x_j) = x_j \partial_j (\|\nabla u\|^2) + \|\nabla u\|^2 (\partial_j x_j)$$

Since the divergence of the position vector field x in d dimensions is $\partial_j x_j = d$, we have

$$x_j \partial_j (\|\nabla u\|^2) = \nabla(\|\nabla u\|^2 x) - d\|\nabla u\|^2$$

Hence

$$-\frac{1}{2} \int_{\Omega} (\nabla(\|\nabla u\|^2 x) - d\|\nabla u\|^2) dx = -\frac{1}{2} \int_{\Omega} \nabla(\|\nabla u\|^2 x) dx - \frac{d}{2} \int_{\Omega} \|\nabla u\|^2 dx$$

From the Divergence theorem

$$\int_{\Omega} \nabla(\|\nabla u\|^2 x) dx = \int_{\partial\Omega} \|u\|^2 \langle x, N \rangle d\sigma$$

So the above evaluates to

$$(\star\star) \quad -\int_{\Omega} \|\nabla u\|^2 dx - \frac{1}{2} \int_{\partial\Omega} \|u\|^2 \langle x, N \rangle d\sigma + \frac{d}{2} \int_{\Omega} \|\nabla u\|^2 dx = \left(\frac{d-2}{2}\right) \int_{\Omega} \|\nabla u\|^2 dx - \frac{1}{2} \int_{\partial\Omega} \|u\|^2 \langle x, N \rangle d\sigma$$

We combine $(\star)$, $(\star\star)$ and we are done. □

Question 4

Let $F(x, y, z) = (x, y, z)$ and let $\Omega \subseteq \mathbb{R}^3$ be a smooth bounded open set. Let N be the outward pointing unit normal and we will calculate

$$\int_{\partial\Omega} N \, d\sigma$$
$$\int_{\partial\Omega} F \wedge N \, d\sigma$$

Solution: By the remark, let $N = (N_1, N_2, N_3)$, since N is outward pointing unit vector we can use the dot product with the standard basis vector $e_1 = (1, 0, 0)$ to obtain that

$$\int_{\partial\Omega} N \, d\sigma = \int_{\partial\Omega} N_1 \, d\sigma = \int_{\partial\Omega} (1, 0, 0) \cdot N \, d\sigma$$

Now we can apply the Divergence theorem for the constant vector field $(1, 0, 0)$, since the divergence is just $\nabla \cdot (1, 0, 0) = 0$ hence

$$\int_{\partial\Omega} (1, 0, 0) \cdot N \, d\sigma = \int_{\Omega} 0 \, dV = 0$$

We can repeat this process for the y and z components using $e_2 = (0, 1, 0)$, $e_3 = (0, 0, 1)$ hence

$$\int_{\partial\Omega} N \, d\sigma = 0$$

For the second part, we start by calculate the cross product of F and N

$$F \times N = (yN_3 - zN_2, zN_1 - xN_3, xN_2 - yN_1)$$

We look at the first component

$$yN_3 - zN_2 = (0, -z, y) \cdot (N_1, N_2, N_3)$$

If we apply the Divergence theorem to the vector field $(0, -z, y)$ which it's div is 0 we will get that the volume integral is zero and the same for the vector fields $(z, 0, -x)$ and $(-y, x, 0)$ for the second and third components hence

$$\int_{\partial\Omega} F \wedge N \, d\sigma = 0$$

□

Question 5

Let $\Omega \subseteq \mathbb{R}^d$ be a smooth domain and let $p \in \partial\Omega$. Let $p \in W \subseteq \mathbb{R}^d$ be an open neighborhood of p . Denote by $B \subseteq \mathbb{R}^d$ the open unit ball and let $\varphi : B \rightarrow W$ be a diffeomorphism such that

$$\begin{aligned}\varphi(0) &= p \\ \varphi(\{x_d > 0\}) &= W \cap \Omega \\ \forall x \in B, \quad \det D\varphi_x &> 0\end{aligned}$$

Subquestion 1

Let $A \in M_d(\mathbb{R})$ be a square matrix such that $\det A > 0$ and let $v_1, \dots, v_{d-1} \in \mathbb{R}^d$ be linearly independent. We will show that

$$\langle A(v_1 \wedge \dots \wedge v_{d-1}), Av_1 \wedge \dots \wedge Av_{d-1} \rangle > 0$$

Proof: By definition,

$$\langle v_1 \times \dots \times v_{d-1}, u \rangle = \det(v_1, \dots, v_{d-1}, u)$$

Let $w = v_1 \times \dots \times v_{d-1}$, $Y = Av_1 \times \dots \times Av_{d-1}$ we aim to show that $\langle Aw, Y \rangle > 0$.

We take $u = Aw$ and obtain that

$$\langle Y, Aw \rangle = \det(Av_1, \dots, Av_{d-1}, Aw) = \det(A) \cdot \det(v_1, \dots, v_{d-1}, w) = \det(A) \langle w, w \rangle = \det(A) \|w\|^2$$

Since $\det(A) > 0$ is given, and $v_1, \dots, v_{d-1}$ are linearly independent hence their cross product w is not the zero vector hence $\|w\|^2 > 0$ and finally we obtain that $\langle Aw, Y \rangle > 0$ as wanted. $\square$

Subquestion 2

We will prove that

$$\langle D\varphi_p^{-1}(\partial_1 \varphi \upharpoonright_p \wedge \dots \wedge \partial_{d-1} \varphi \upharpoonright_p), e_d \rangle > 0$$

Proof: We use the previous subquestion by choosing $A = (D\varphi)_p^{-1} = (D\varphi_0)^{-1}$ and $v_i = \partial_i \varphi \upharpoonright_p = D\varphi_0(e_i)$ for $1 \leq i \leq d-1$. Since $\det D\varphi_x > 0$ for all $x \in B$, we have $\det D\varphi_0 > 0$, which implies

$$\det A = \det(D\varphi_0^{-1}) = \frac{1}{\det D\varphi_0} > 0$$

The vectors $v_1, \dots, v_{d-1}$ are linearly independent because φ is a diffeomorphism so from the previous subquestion

$$\langle D\varphi_p^{-1}(\partial_1 \varphi \upharpoonright_p \wedge \dots \wedge \partial_{d-1} \varphi \upharpoonright_p), D\varphi_p^{-1}(\partial_1 \varphi \upharpoonright_p) \wedge \dots \wedge D\varphi_p^{-1}(\partial_{d-1} \varphi \upharpoonright_p) \rangle > 0$$

We have $D\varphi_p^{-1}(\partial_i \varphi \upharpoonright_p) = D\varphi_0^{-1}(D\varphi_0(e_i)) = e_i$ and substituting this into the right hand side of the inner product gives

$$\langle D\varphi_p^{-1}(\partial_1 \varphi \upharpoonright_p \wedge \dots \wedge \partial_{d-1} \varphi \upharpoonright_p), e_1 \wedge \dots \wedge e_{d-1} \rangle > 0$$

By definition, for any vector u , we have $\langle e_1 \wedge \dots \wedge e_{d-1}, u \rangle = \det(e_1, \dots, e_{d-1}, u)$. Let $u = e_d$ and we will get

$$\det(e_1, \dots, e_{d-1}, e_d) = \det(I_d) = 1$$

Since $e_1 \wedge \dots \wedge e_{d-1}$ is orthogonal to $e_1, \dots, e_{d-1}$, it points purely in the e_d direction, meaning $e_1 \wedge \dots \wedge e_{d-1} = e_d$, substituting this back

$$\langle D\varphi_p^{-1}(\partial_1 \varphi \upharpoonright_p \wedge \dots \wedge \partial_{d-1} \varphi \upharpoonright_p), e_d \rangle > 0$$

$\square$

Subquestion 3

Denote $N = \partial_1 \varphi \upharpoonright_p \wedge \dots \wedge \partial_{d-1} \varphi \upharpoonright_p$. We will prove that there exists $\delta > 0$ such that for all $t \in (0, \delta)$ we have $p - tN \notin \Omega$.

Proof: The diffeomorphism φ maps the upper half-ball region $\{x \in B \mid x_d > 0\}$ into the interior domain $W \cap \Omega$, which means that a point $y \in W$ belongs to Ω if and only if the d -th component of its preimage is strictly positive ($y \in \Omega \iff \langle \varphi^{-1}(y), e_d \rangle > 0$).

We look at the path $\gamma(t) = p - tN$. Since W is an open neighborhood of p and $\gamma(0) = p$, by continuity there exists $\delta_1 > 0$ such that $\gamma(t) \in W$ for all $t \in [0, \delta_1)$.

Using Taylor Expansion around $t = 0$ we obtain that

$$\varphi^{-1}(p - tN) = \varphi^{-1}(p) + D\varphi_p^{-1}(-tN) + o(t) = -tD\varphi_p^{-1}(N) + o(t)$$

Taking the inner product with e_d to isolate the d -th component

$$\langle \varphi^{-1}(p - tN), e_d \rangle = -t\langle D\varphi_p^{-1}(N), e_d \rangle + \langle o(t), e_d \rangle$$

From the second subquestion, we know that $\langle D\varphi_p^{-1}(N), e_d \rangle = C$ for some strictly positive constant $C > 0$. Therefore

$$\langle \varphi^{-1}(p - tN), e_d \rangle = -tC + o(t) = t\left(-C + \frac{o(t)}{t}\right)$$

Since $\lim_{t \rightarrow 0} \frac{o(t)}{t} = 0$, there exists a sufficiently small $\delta \in (0, \delta_1)$ such that for all $t \in (0, \delta)$, we have $|\frac{o(t)}{t}| < C$, so that $-C + \frac{o(t)}{t} < 0$. Thus, for all $t \in (0, \delta)$ we have $\langle \varphi^{-1}(p - tN), e_d \rangle < 0$ and this implies $p - tN \notin \Omega$. $\square$